GEL-Ped: An instability-triggered safeguard for short-horizon pedestrian prediction under changing crowd flows

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Abstract

Safe navigation around pedestrians depends on forecasts that arrive soon enough to change the next action of a robot, vehicle, or crowd-management system. Learned interaction models can be accurate in familiar conditions yet issue unstable corrections when streams cross, merge, or enter through an unfamiliar layout. Existing work offers strong predictors and warning scores, but a warning alone does not say which forecast should be used now. This paper presents GEL-Ped (Guarded Ensemble Learning for Pedestrians), a runtime safeguard for short-horizon prediction. Independently trained graph models provide the default forecast. GEL-Ped monitors their local disagreement, compares it with an envelope learned from complete training runs, and leaves the graph forecast unchanged while disagreement remains inside that envelope. An exceedance moves the forecast smoothly toward a feature-based fallback with an explicit speed cap. Complete-run experiments separate ordinary accuracy from tail failures, test whether routing is genuinely targeted, repeat the analysis across retrained ensembles and a second neural backbone, and finish with a one-shot untouched topology shift. The evidence supports a focused conclusion: disagreement can temper member-specific neural instability without replacing the primary predictor, but it cannot repair errors shared by the entire model family.

Keywords: pedestrian prediction, runtime assurance, graph neural network, deep ensemble, distribution shift, tail risk

1. Introduction

A pedestrian forecast has operational value only if it arrives early enough to change a decision. A mobile robot may slow before entering a person's path, an automated vehicle may delay a turn, and a station-monitoring system may flag a developing conflict. These applications repeatedly need a short response forecast from motion observed up to the present. This task differs from predicting a final destination or several multi-second routes: current velocity, nearby interactions, fast updates, and graceful behaviour outside familiar conditions matter most (Rudenko et al., 2020; Kothari et al., 2022).

Crowds make even this short horizon difficult. People preserve personal space, anticipate encounters, and change speed or passing side when local flows meet (Moussaïd et al., 2011; Karamouzas et al., 2014). Learned interaction models represent such coupled responses well on average. An average score, however, is not an operating rule. A rare unstable correction can matter more to the next control action than thousands of routine forecasts. An uncertainty score may warn that a case is unusual, but it does not itself provide a replacement forecast. A permanent hybrid supplies a forecast but constrains the learned model even when it works normally. Online adaptation can recover after feedback, although it changes parameters or weights and cannot act before suitable errors arrive (Ivanovic et al., 2023; Park et al., 2024).

This leaves a practical gap between detection and action. A useful safeguard must answer four linked questions: can an unreliable correction be detected before its future error is known; what forecast should replace or temper it; is the action targeted to the flagged cases rather than benefiting from a generally stronger fallback; and which errors remain invisible? Uncertainty estimation, selective prediction, online experts, and runtime assurance address parts of this chain (Ovadia et al., 2019; Malinin et al., 2021; Geifman and El-Yaniv, 2017; Sha, 2001). They are less often evaluated together as one causal mechanism for repeated pedestrian prediction.

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GEL-Ped joins these pieces with a small, auditable rule. Independently trained graph predictors estimate corrections around recent velocity, and their mean is the default forecast. The method measures how strongly they disagree for the same pedestrian, using only past and present observations. A complete-run calibration envelope defines ordinary disagreement. Inside the envelope the graph retains full control. Only an exceedance reduces its weight, moving the output toward a compact histogram-gradient-boosting (HGB) fallback built from recent motion, personal space, forward attention, and closing motion. The method does not classify a facility or claim that every distribution shift is detected. It asks a narrower question: are plausible versions of the deployed neural predictor disagreeing more than they did during supported operation?

That question also defines the method's boundary. Disagreement can reveal a *variance-type failure*, in which trained members react differently to the same situation. It cannot reveal a *shared-bias failure*, in which all members agree on the same wrong correction. Richer context, broader training data, a more diverse model family, or adaptation may be needed for the latter. GEL-Ped therefore returns a point forecast and an intervention level, not a probability of error or a safety certificate.

The paper makes three contributions. First, it introduces an identity-preserving guard: the neural forecast is unchanged inside a leakage-free calibration envelope and is tempered only after a local exceedance. Second, it evaluates that action using complete physical runs, tail endpoints, targeting controls, retraining variation, equivalence tests, and a one-shot untouched topology shift. Third, it quantifies the operating boundary between member-specific instability, which disagreement can expose, and shared model-family error, which it cannot.

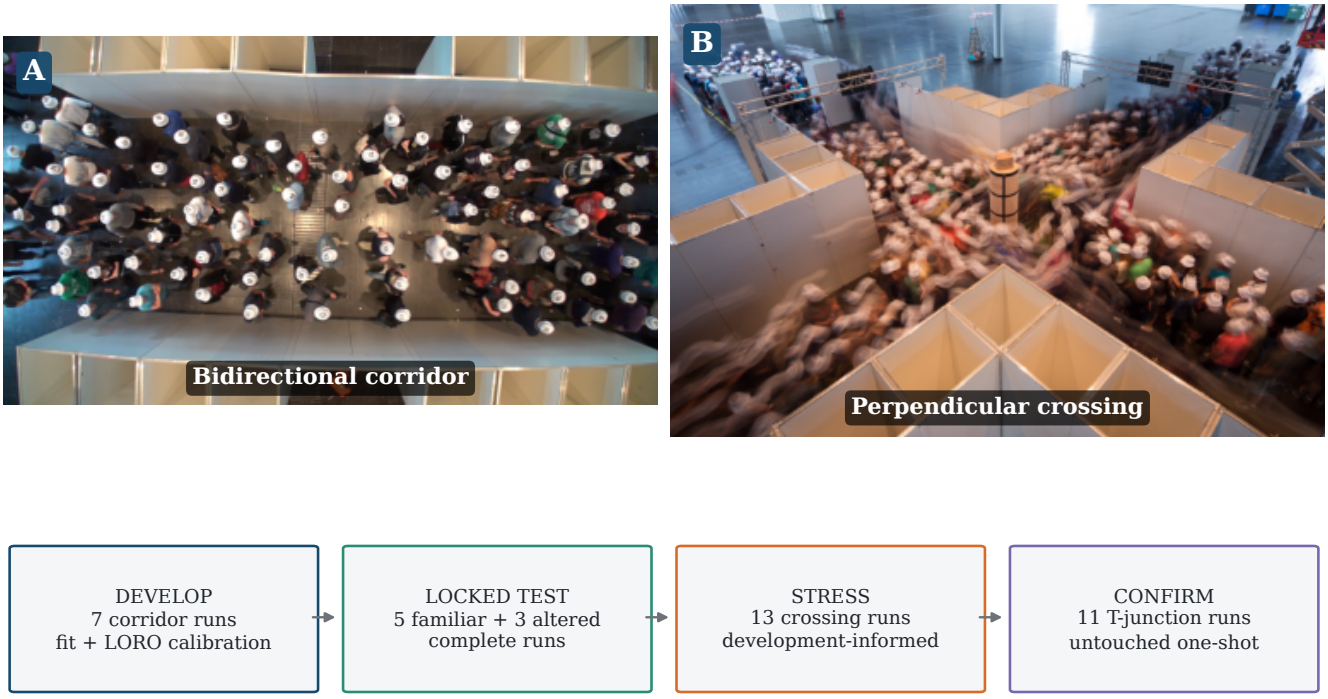


Figure 1: Empirical settings and frozen chronology. Complete physical runs, rather than shuffled frames, define fitting, calibration, stress evaluation, and untouched confirmation. The crossing archive informed mechanism development; the T-junction archive remained unopened for performance analysis until the method, endpoints, code, and multiplicity plan were frozen. Photographs: Juelich Pedestrian Dynamics Data Archive, CC BY 4.0; DOIs 10.34735/ped.2013.5 and 10.34735/ped.2013.4.

2. Related work and research gap

2.1. Pedestrian interaction and task scope

Pedestrian forecasting spans the next movement response through long-horizon multimodal paths. Social-force, collision-prediction, and velocity-obstacle models express persistence and encounter geometry with interpretable variables (Helbing and Molnár, 1995; Zanlungo et al., 2011; Karamouzas et al., 2014; van den Berg et al., 2008). Learned predictors use recurrent pooling, graphs, attention, and multimodal decoders to represent richer interactions (Alahi et al., 2016; Gupta et al., 2018; Mohamed et al., 2020; Salzmann et al., 2020). Surveys caution that results do not transfer automatically across histories, horizons, sampling rates,

and output types (Rudenko et al., 2020; Kothari et al., 2022). Empirical crowd research likewise distinguishes experimental control from validity for a target crowd and facility (Haghani and Sarvi, 2018; Haghani, 2023). Published multi-second errors are therefore not matched baselines for the present 0.8-s-history, 0.4-s-response task.

Recent transport studies add structure that reflects the operating setting. Learned latent modes can make pedestrian interaction predictions more interpretable (Liu et al., 2024); visual and contextual cues can connect crossing intention with trajectory prediction (Munir and Kucner, 2025); and a utility layer can select among learned cyclist trajectories in rare interactions (Li et al., 2023). At the aggregate level, an interpretable pedestrian-flow model can be recalibrated online as a facility changes (Mai et al., 2025). These studies improve representation or adaptation. GEL-Ped addresses the immediate decision that precedes either: what point forecast should a fixed short-horizon predictor return when its own members become unstable?

2.2. From warning scores to a usable forecast

Uncertainty estimates can deteriorate as test data move away from training conditions (Ovadia et al., 2019). The Shifts benchmark extends this concern to real distribution changes and promotes risk–retention evaluation, including for motion prediction (Malinin et al., 2021). Mahalanobis distance measures how unusual an input representation is relative to training data (Lee et al., 2018); ensemble disagreement asks whether plausible trained models give different answers. Ensemble-based epistemic uncertainty has supported novelty detection and recovery in autonomous driving (Filos et al., 2020). Neither score is automatically the probability that a pedestrian forecast is wrong.

Selective prediction rejects cases above a threshold and measures the trade-off between retained coverage and risk (Geifman and El-Yaniv, 2017; Cortes et al., 2016). A local planner may still require a trajectory at every update. Adaptation and online expert methods instead learn from delayed errors (Herbster and Warmuth, 1998; Cesa-Bianchi and Lugosi, 2006; Ivanovic et al., 2023; Park et al., 2024). Adaptive conformal methods update uncertainty regions as observations arrive (Gibbs and Candès, 2021; Dixit et al., 2023). These methods complement GEL-Ped: they learn after feedback or construct sets, whereas GEL-Ped provides an immediate point forecast without updating the primary model.

Runtime-assurance architectures supervise a capable component with a simpler response when conditions become unacceptable (Sha, 2001). Switching or weighting is therefore not itself new. The unresolved pedestrian-specific issue is whether a causally available local signal can identify the occasions on which a compact fallback is useful, beyond fixed weighting, generic input distance, or delayed expert updates.

Table 1: Positioning by deployment question. Detection and action are separated because a warning score alone does not produce a pedestrian forecast.

Approach	Detection signal	Action after detection	Scope and use
Shift monitoring	confidence, feature distance, or disagreement	score, retain, recover, or request feedback	identifies conditions in which ordinary confidence may be unreliable
Selective prediction	confidence threshold	reject or abstain	trades coverage for lower retained risk
Adaptation and online experts	delayed errors or test-time information	update sets, parameters, or expert weights	recovers after feedback in a changing stream
GEL-Ped	local disagreement relative to a complete-run envelope	continuously temper the graph forecast with a feature-based fallback	supplies an immediate forecast and intervention level

The gap is the complete causal chain. The monitor must detect the target failure before its label arrives; the action must still return a forecast; matched controls must distinguish useful targeting from a generally good fallback; and the monitor's blind spots must be explicit. GEL-Ped is designed and evaluated around that chain.

3. GEL-Ped method

3.1. Prediction task and design logic

For pedestrian i at forecast origin t, inputs contain positions no later than t. The target is mean velocity over the next $\Delta = 0.4$ s. Recent velocity over $H_o = 0.8$ s provides an interpretable anchor,

$$\mathbf{b}_i(t) = \frac{\mathbf{x}_i(t) - \mathbf{x}_i(t - H_o)}{H_o}, \qquad \mathbf{v}_i^+(t) = \frac{\mathbf{x}_i(t + \Delta) - \mathbf{x}_i(t)}{\Delta}. \tag{1}$$

A past-only route unit vector $\mathbf{e}_i$ defines the forward axis and $\boldsymbol{\ell}_i = (-e_{iy}, e_{ix})$ the lateral axis. The primary analysis uses prior velocity; an entrance-derived axis is only a sensitivity. Wall fields are excluded from primary models so a layout-specific boundary code cannot define transfer.

The three components follow directly from the deployment problem. The graph ensemble is used because interactions are relational and independent fits expose training sensitivity. The fallback uses compact aggregate cues so that its errors need not mirror the graph's message passing. The gate is one-sided because routine neural forecasts should not be altered merely to obtain a small ensemble gain.

Table 2: Notation. Every quantity used by the deployed predictor is computed from times no later than t.

Symbol	Meaning	Role
$\mathbf{x}_i(t)$	planar position	observed through t
$\mathbf{b}_i(t)$	recent-velocity anchor	displacement over $H_o = 0.8$ s
$\mathbf{v}_i^+(t)$	future mean velocity target	offline evaluation only
$\mathbf{e}_i, \boldsymbol{\ell}_i$	past-only forward and lateral axes	local coordinates
$\widehat{\mathbf{v}}_i^{G,m}$	graph-member prediction	$M = 5$ in the primary ensemble
$\widehat{\mathbf{v}}_i^F$	feature-based fallback prediction	same anchor and focal inputs
$s_i(t), S_i(t)$	instantaneous and smoothed disagreement	local to one pedestrian
τ, q_{IQR}	envelope threshold and scale	out-of-fold complete-run estimates
$g_i(t)$	graph weight	one inside the envelope

3.2. Causal interaction representation

At most eight neighbours within 3 m are retained. For neighbour j, let $\mathbf{r}_{ij} = \mathbf{x}_j - \mathbf{x}_i$, $d_{ij} = \|\mathbf{r}_{ij}\|$, and $\mathbf{u}_{ij} = \mathbf{b}_j - \mathbf{b}_i$. With body radius $r_b = 0.25$ m and interaction range $\lambda = 0.8$ m, proximity is weighted by

$$m_{ij} = \exp\left[-\frac{d_{ij} - 2r_b}{\lambda}\right]. \tag{2}$$

Three vector summaries describe nearby people: isotropic personal-space pressure, forward-weighted pressure, and closing motion. The forward term weights neighbour j by $[\max(\mathbf{r}_{ij}^{\mathsf{T}}\mathbf{e}_i/d_{ij}, 0)]^2$; the closing term also weights approach speed, capped at 3 m/s. Recent velocity, route direction, and these fields are projected onto $(\mathbf{e}_i, \boldsymbol{\ell}_i)$, yielding ten focal scalars. Each retained neighbour contributes local position, relative velocity, distance, closing speed, and a mask. These quantities are behavioural cues, not a complete behavioural theory.

3.3. Graph and fallback experts

Each graph member transforms the seven neighbour variables with a shared $7 \to 48 \to 48$ encoder, attention-pools valid neighbours, and joins that message with the ten focal scalars. A $58 \to 96 \to 48 \to 2$ head predicts forward and lateral corrections,

$$\widehat{\mathbf{v}}_i^{G,m} = \mathbf{b}_i + \delta_{i,\parallel}^{G,m}\mathbf{e}_i + \delta_{i,\perp}^{G,m}\boldsymbol{\ell}_i, \qquad m = 1, \ldots, M. \tag{3}$$

The graph mean is $\bar{\mathbf{v}}_i^G = M^{-1}\sum_m \widehat{\mathbf{v}}_i^{G,m}$. Each member has 13,203 trainable parameters. Runs contribute equally to the squared-error loss, irrespective of their number of origins. Members use AdamW for 25 epochs, learning rate 10^{-3}, weight decay 10^{-3}, and gradient-norm cap 5.

The fallback sees the ten focal scalars but no individual-neighbour messages. Separate HGB regressors estimate the two residual components,

$$\widehat{\mathbf{v}}_i^F = \mathbf{b}_i + \delta_{i,\parallel}^F\mathbf{e}_i + \delta_{i,\perp}^F\boldsymbol{\ell}_i. \tag{4}$$

Each uses 200 iterations, at most 31 leaves, minimum leaf size 50, learning rate 0.05, and L_2 regularisation 1. Both experts apply the same 3 m/s output-speed cap. The fallback is selected for bounded piecewise-constant extrapolation and a different inductive bias, not because it is always more accurate.

3.4. Local instability guard

The instantaneous disagreement for one pedestrian is

$$s_i(t) = \left[\frac{1}{M} \sum_{m=1}^{M} \left\| \hat{\mathbf{v}}_i^{G,m}(t) - \bar{\mathbf{v}}_i^{G}(t) \right\|_2^2 \right]^{1/2}. \tag{5}$$

Forecast origins are 0.4 s apart. The smoothed score $S_i(t)$ averages the current and up to four preceding scores for the same pedestrian within a trailing 1.6-s window. State resets between pedestrians and scenes.

Calibration is leave-one-complete-run-out. Models fit six runs and score the seventh chronologically. Pooling only out-of-fold scores gives $S_{(1)} \le \cdots \le S_{(N)}$. The threshold is the finite-sample 90th percentile,

$$\tau = S_{(k)}, \qquad k = \min\{N, \lceil (N+1)0.90 \rceil\}, \tag{6}$$

and $q_{\mathrm{IQR}} = \max\{Q_{0.75} - Q_{0.25}, 10^{-6}\}$ m/s. The deployed weight and prediction are

$$g_i(t) = \exp\left[-\frac{(S_i(t) - \tau)_+}{q_{\mathrm{IQR}}} \right], \qquad \hat{\mathbf{v}}_i(t) = g_i(t)\bar{\mathbf{v}}_i^{G}(t) + (1 - g_i(t))\hat{\mathbf{v}}_i^{F}(t). \tag{7}$$

Thus $g_i(t) = 1$ inside the envelope. The fallback has no compulsory weight. Linear and logistic maps are sensitivities; the exponential rule is frozen primary. Figure 2 shows the calculation and cross-fitting loop.

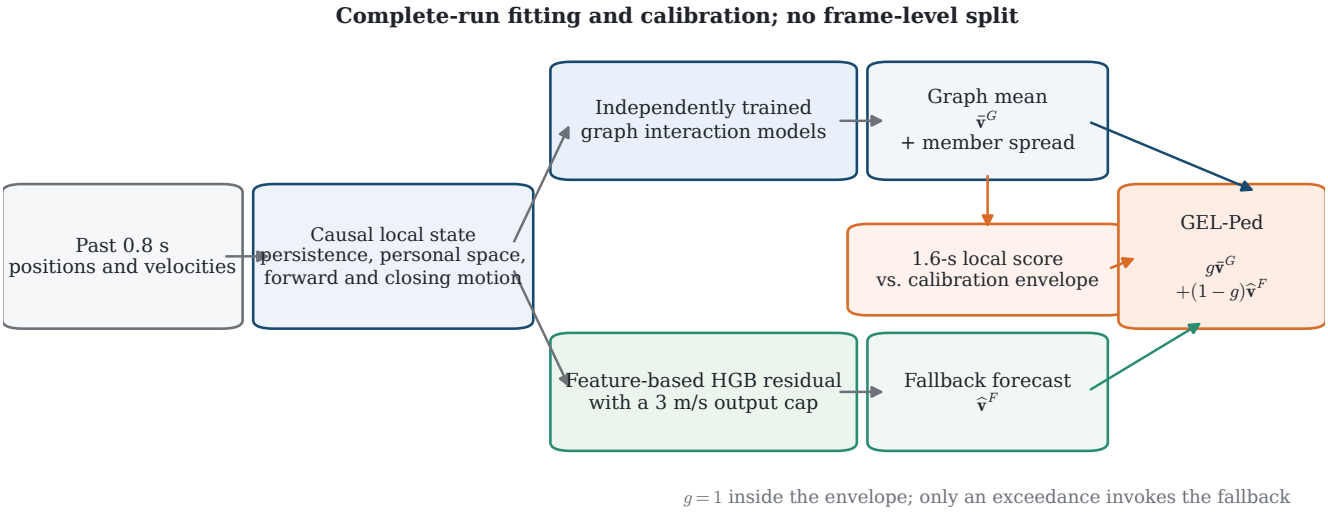


Figure 2: GEL-Ped. Five anchored graph learners provide the default mean and a local disagreement score. Leave-one-complete-run-out calibration fixes the envelope. Only an exceedance moves the forecast toward the feature-based HGB fallback. All deployed signals use past or present observations.

4. Experimental design

4.1. Data roles and one-shot firewall

The controlled archive contains 25-Hz bidirectional-corridor and right-angle-crossing trajectories (Boltes and Seyfried, 2013; Boltes et al., 2020). Seven complete corridor runs form the fitting and calibration set. Five locked tests retain the 22-m corridor and 5-m exit; three alter corridor length and/or exit arrangement. Thirteen right-angle-crossing runs informed mechanism development and are treated as a stress set, not as untouched confirmation.

The one-shot confirmation uses 11 runs from a separately archived 16-Hz Juelich T-junction experiment (DOI 10.26165/JUELICH-DATA/YNCHYF). Before labelled performance inspection, the archive digest, inclusion rules, task, models, four endpoints, multiplicity correction, and analysis code were frozen. Tracks are projected to 10 Hz using only the two most recent source observations at each grid time. No architecture, threshold, feature, cap, or endpoint was changed after the first result was opened.

Origins require the full 0.8-s history and 0.4-s target for offline scoring. A 3 m/s quality limit is applied only to speed observed in the history. Model fitting samples at most 6,000 eligible origins per run with a fixed seed; evaluation uses every eligible origin in chronological order. The deployed predictor never receives a future position, regime label, or test error.

4.2. Comparators and replication

Primary comparators are constant velocity, one graph member, the graph mean, the coordinate-wise member median, and the HGB fallback. Routing controls share origins, graph predictions, fallback predictions, and member draws: a calibration-selected fixed blend; within-run permutation of g; Mahalanobis, graph–fallback disagreement, learned, and speed gates; and a causal delayed-error expert weight. The latter updates only after the corresponding 0.4-s target becomes observable. A future-error oracle is an unattainable ceiling, not a deployable baseline.

The primary ensemble has five members. A pre-trained pool of 25 independent seeds supports 30 member draws and ensemble sizes $M \in \{3, 5, 10, 20\}$, with τ and q_{IQR} recalibrated for each draw. A compact interaction Transformer replaces the graph in a backbone check while retaining the task, inputs, splits, fallback, and guard.

4.3. Endpoints and inference

Vector RMSE is computed within each physical run and macro-averaged. Tail endpoints are the 90th, 95th, and 99th percentiles, CVaR at 95% and 99%, and the rate above 0.5 m/s (0.2 m displacement over the prediction interval). Detection of the largest 5% of graph errors uses AUROC, AUPRC, and risk–retention area. Complete-run bootstrap intervals resample runs; retraining intervals cross complete runs with member draws. Practical equivalence to the graph is assessed at ± 0.002 m/s and $\pm 1\%$ RMSE. The four one-shot endpoints use exact paired sign tests with Holm correction.

The intervention stratum is $(S_i(t) - \tau)/q_{\mathrm{IQR}}$; values above two are more than two calibration IQRs beyond the envelope. Decomposition uses the identity member MSE = squared error of the ensemble mean + between-member variance. Operational consequences are checked in paired, open-loop robot replays over recorded crowds. Pedestrians do not react to the robot, so this is a consequence audit rather than a reciprocal simulation or safety proof.

5. Results

5.1. The warning is informative, but not a general error detector

In the crossing stress set, spread detects the largest 5% of graph errors with mean AUROC 0.652 and AUPRC 0.087. Graph–HGB disagreement is slightly stronger (0.672 and 0.097), while speed (0.525 and 0.058) and Mahalanobis distance (0.564 and 0.066) are weaker. The same ordering appears in risk–retention area: lower is better, and crossing values are 0.110 for graph–HGB disagreement, 0.114 for spread, 0.130 for Mahalanobis distance, and 0.143 for speed. Spread therefore contains useful local information, but it is neither an error probability nor the best possible detector.

The partial Spearman association between spread and graph error after conditioning on observed speed averages 0.17 across crossing runs. The signal is not simply a speed threshold. Its calibrated activation is strongly regime dependent: the deployed graph exceeds τ on 2.0% of familiar, 3.6% of altered, and 40.5% of crossing forecasts. The first origin of a new familiar track fires more often (10.4%) before the temporal average stabilises, a cold-start behaviour that downstream systems should retain rather than hide.

5.2. In-support equivalence and crossing-tail control

In the primary deployed ensemble, familiar RMSE changes from 0.173532 to 0.173655 m/s and altered-geometry RMSE from 0.168381 to 0.168404 m/s. Both meet the two pre-stated equivalence margins. The HGB fallback is materially worse in these regimes, so replacing the graph globally is not supported.

In crossing, GEL-Ped changes RMSE from 0.3761 to 0.3687 m/s (2.0%), $q_{.95}$ from 0.6839 to 0.6684 m/s (2.3%), $\mathrm{CVaR}_{.95}$ from 0.8479 to 0.8326 m/s (1.8%), and the error rate above 0.5 m/s from 15.78% to 14.72%. HGB and delayed-error weighting achieve lower crossing averages, but HGB loses clearly in the corridor regimes and the online gate depends on labels that arrive later. GEL-Ped's value is immediate, label-free routing while the learned expert is unstable, not universal superiority over either expert.

Table 3: Run-macro mean and tail error for GEL-Ped and matched controls. All methods use the same evaluation origins; gates use the same graph–fallback expert pair.

Regime	Method	RMSE	$q_{.95}$	$\mathrm{CVaR}_{.95}$	$e > 0.5$
altered geometry	Graph ensemble	0.1684	0.3135	0.3846	0.36%
altered geometry	GEL-Ped	0.1684	0.3134	0.3844	0.36%
altered geometry	HGB fallback	0.1903	0.3496	0.4214	0.59%
altered geometry	Fixed blend	0.1684	0.3131	0.3844	0.36%
altered geometry	Learned matched gate	0.1710	0.3163	0.3874	0.36%
altered geometry	Delayed-error gate	0.1685	0.3135	0.3847	0.36%
crossing topology	Graph ensemble	0.3761	0.6839	0.8479	15.78%
crossing topology	GEL-Ped	0.3687	0.6684	0.8326	14.72%
crossing topology	HGB fallback	0.3646	0.6602	0.8294	13.98%
crossing topology	Fixed blend	0.3747	0.6810	0.8452	15.59%
crossing topology	Learned matched gate	0.3664	0.6637	0.8296	14.36%
crossing topology	Delayed-error gate	0.3636	0.6580	0.8264	13.90%
familiar corridor	Graph ensemble	0.1735	0.3192	0.3961	0.50%
familiar corridor	GEL-Ped	0.1737	0.3194	0.3963	0.50%
familiar corridor	HGB fallback	0.2004	0.3649	0.4445	0.88%
familiar corridor	Fixed blend	0.1736	0.3196	0.3960	0.50%
familiar corridor	Learned matched gate	0.1773	0.3248	0.4012	0.52%
familiar corridor	Delayed-error gate	0.1738	0.3200	0.3967	0.51%

Velocity-error quantities are in m/s. The final column is a forecast share.

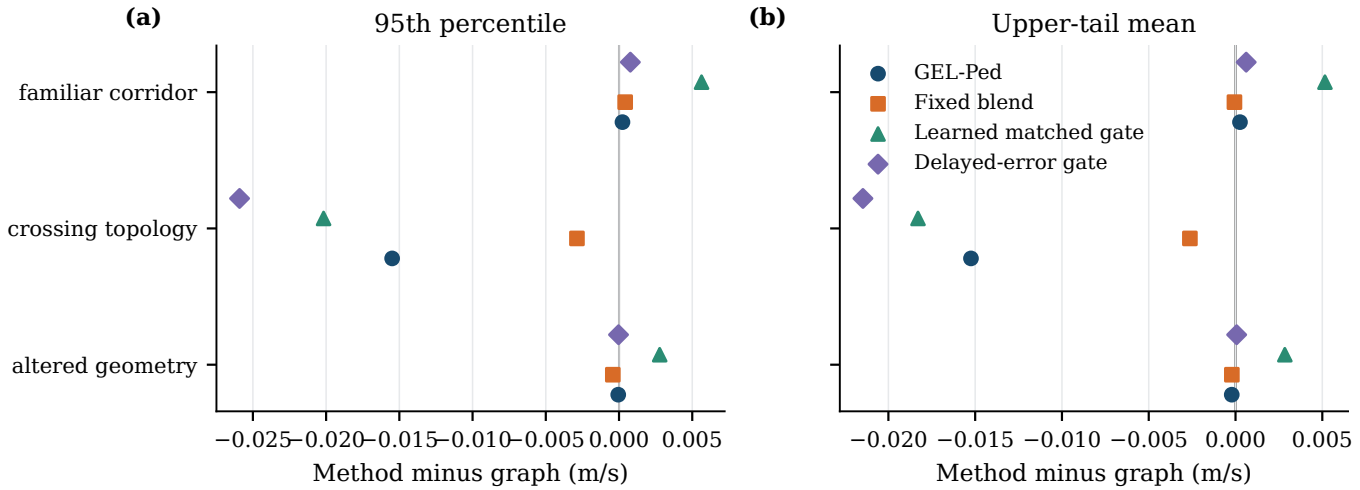


Figure 3: Mean and tail changes relative to the direct graph ensemble. Each point is a complete-run macro difference under a matched routing rule; negative values favour the comparison method.

Table 4: Where the guard acts. Strata are defined by calibration-normalised ensemble spread; permutation gaps compare the observed routing with the same weights randomly reassigned within a run.

Regime	Spread stratum	Share	Graph RMSE	GEL-Ped RMSE	Permutation gap	Variance share
altered geometry	inside	96.4%	0.1668	0.1668	0.0001	3.5%
altered geometry	0–1 IQR	3.2%	0.2062	0.2062	0.0001	5.7%
altered geometry	1–2 IQR	0.3%	0.2336	0.2342	-0.0006	7.7%
altered geometry	>2 IQR	0.1%	0.2293	0.2235	0.0055	12.9%
crossing topology	inside	59.5%	0.3434	0.3434	-0.0031	1.2%
crossing topology	0–1 IQR	24.0%	0.4035	0.3944	0.0041	1.7%
crossing topology	1–2 IQR	9.3%	0.4393	0.4204	0.0119	2.2%
crossing topology	>2 IQR	7.2%	0.4760	0.4554	0.0127	5.7%
familiar corridor	inside	98.0%	0.1721	0.1721	0.0001	2.9%
familiar corridor	0–1 IQR	1.8%	0.2321	0.2353	-0.0031	4.6%
familiar corridor	1–2 IQR	0.2%	0.2176	0.2310	-0.0134	8.7%
familiar corridor	>2 IQR	0.1%	0.2466	0.2294	0.0172	15.8%

Positive permutation gaps favour targeted routing. Variance share is between-member variance divided by mean member MSE.

5.3. *Targeting, not only favourable blending*

The high-instability stratum contains 7.2% of crossing forecasts. There graph RMSE is 0.4760 m/s and GEL-Ped RMSE is 0.4554 m/s. Randomly reassigning the same run-wise weights worsens RMSE by 0.0127 m/s relative to the observed assignment. A fixed calibration blend is also much closer to the graph on the full crossing set (0.3747 m/s) than GEL-Ped (0.3687 m/s). The benefit is therefore not explained by giving HGB a constant small weight.

The decomposition also shows why the improvement is bounded. In the $>$ 2-IQR crossing stratum, between-member variance is 0.0140 $(\mathrm{m/s})^2$ but squared error shared by the ensemble mean is 0.2303 $(\mathrm{m/s})^2$; variance is only 5.7% of mean member MSE. Inside the envelope it is 1.2%. Most crossing error is shared bias that a same-family disagreement signal cannot see. GEL-Ped reduces a targeted instability component rather than solving the topology shift.

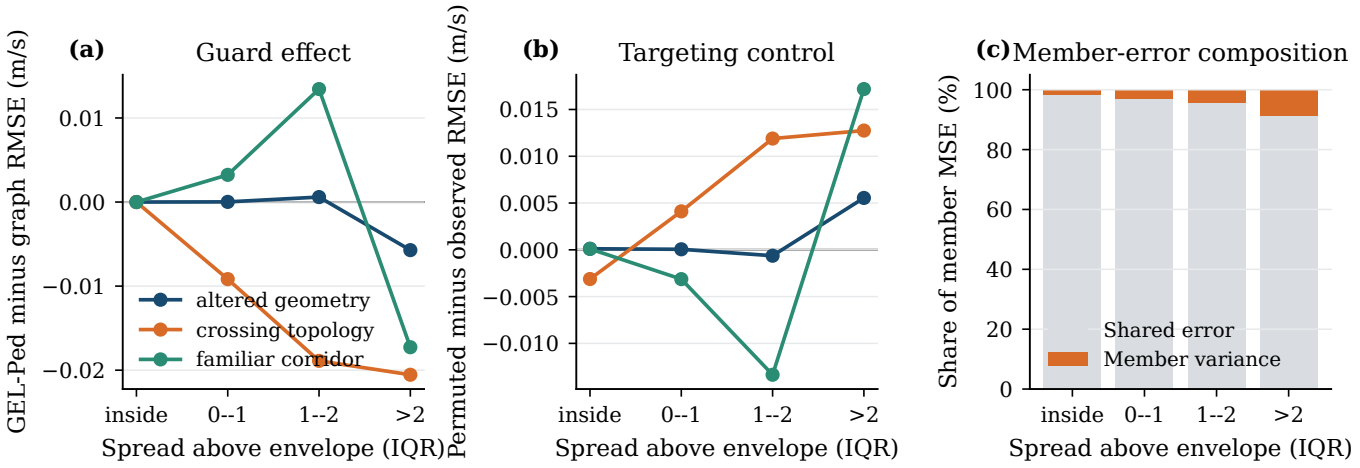


Figure 4: Targeting and decomposition across calibration-normalised spread strata. The permutation control preserves run-wise guard weights but breaks their alignment with individual forecasts.

5.4. *The effect survives retraining and a second backbone*

Across 30 independently drawn five-member ensembles, every draw improves crossing RMSE. The crossed run–draw mean change is -0.0080 m/s with 95% interval $[-0.0117, -0.0049]$; changes in $q_{.95}$ and $\mathrm{CVaR}_{.95}$ are -0.0168 and -0.0176 m/s. Familiar and altered changes remain small ($+0.0004$ m/s in each). The direction persists for $M = 3, 5, 10$, and 20, while between-draw variation shrinks as M grows (Appendix B). This removes the dependence of the earlier result on one selected training seed.

The compact Transformer supplies a separate training recipe and encoder. Guarding changes its crossing RMSE from 0.3767 to 0.3699 m/s (1.8%), while changing familiar RMSE from 0.1888 to 0.1891 m/s and altered RMSE from 0.1841 to 0.1842 m/s. The mechanism is therefore not unique to the graph architecture, although evidence from two compact backbones does not establish architecture-independent performance.

The causal delayed-error gate reaches 0.3636 m/s in crossing, below GEL-Ped's 0.3687 m/s, and is also stronger after enough labels arrive. GEL-Ped remains useful before such feedback and does not update the deployed experts. A system with reliable delayed labels can therefore treat disagreement as an immediate safeguard and online weighting as a complementary adaptation layer.

5.5. *One-shot confirmation on an untouched topology*

The untouched T-junction is the decisive generalisation test. GEL-Ped improves all four locked endpoints across 11 complete runs. RMSE falls by 3.1%, $q_{.95}$ and $\mathrm{CVaR}_{.95}$ by 3.3%, and the large-error rate by 7.1% under the pre-specified run-macro estimand. Bootstrap intervals exclude zero for all four endpoints. After Holm adjustment, $q_{.95}$ ($p = 0.041$) and $\mathrm{CVaR}_{.95}$ ($p = 0.008$) remain significant; RMSE and the large-error rate each have $p = 0.055$. The stronger tail evidence matches the method's intended role.

5.6. *Calibration, computation, and downstream consequence*

The pooled calibration estimates are $\tau = 0.0443$ m/s and $q_{\mathrm{IQR}} = 0.0120$ m/s. A 2,000-replicate complete-run bootstrap gives 95% intervals $[0.0346, 0.0471]$ and $[0.0077, 0.0132]$ m/s. Changing the smoothing window

Table 5: One-shot untouched T-junction evaluation. Inference is paired by complete physical run and adjusted across the four locked endpoints.

Endpoint	Graph	GEL-Ped	Δ	95% CI	Relative Δ	Holm p
RMSE (m/s)	0.3262	0.3141	-0.0121	[-0.0211, -0.0042]	-3.1%	0.055
$q_{.95}$ (m/s)	0.6189	0.5942	-0.0247	[-0.0412, -0.0085]	-3.3%	0.041
$\mathrm{CVaR}_{.95}$ (m/s)	0.7281	0.7004	-0.0277	[-0.0461, -0.0105]	-3.3%	0.008
$e > 0.5$ (share)	0.1192	0.1043	-0.0149	[-0.0281, -0.0043]	-7.1%	0.055

Negative differences favour GEL-Ped. The archive, inclusion rules, models, guard, and endpoints were frozen before labelled inspection.

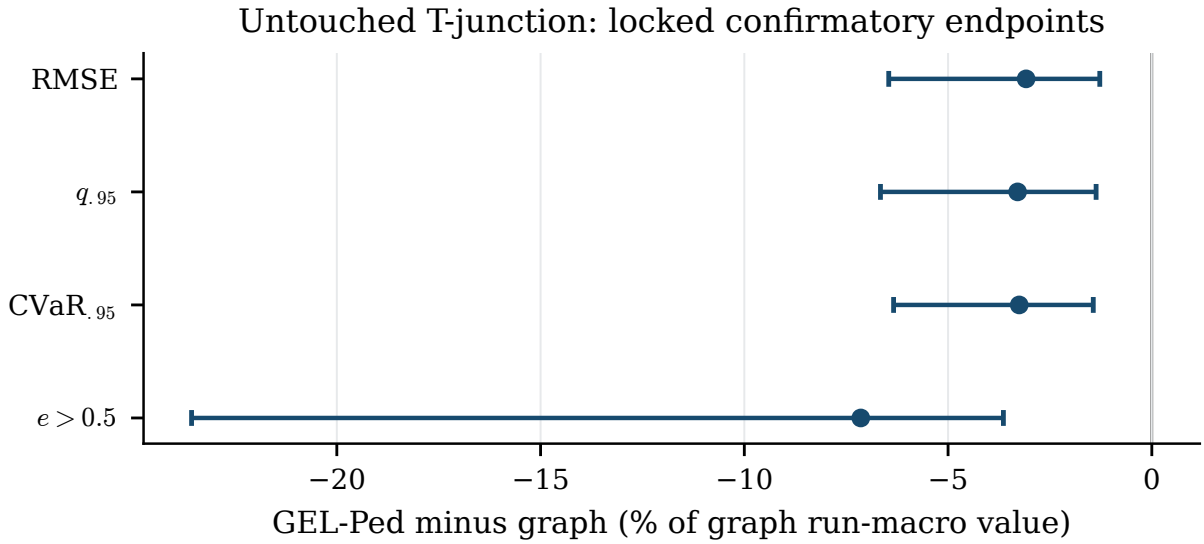


Figure 5: Locked effects in the one-shot untouched T-junction. Intervals resample complete physical runs; effects are normalised only for visual comparison across endpoint units.

from one origin to 0.8, 1.6, or 3.6 s, or replacing the exponential map by a linear or logistic map, changes crossing RMSE by less than 0.004 m/s and does not change the interpretation.

Post-prediction CPU timing is small: the five-member reduction has median 0.006 ms and the guard/blend 0.028 ms; these numbers exclude neural and HGB forward passes. In the paired replay, merely replacing graph forecasts by GEL-Ped under the same clearance rule changes almost no crossing decisions. A naive margin proportional to intervention increases crossing travel time and intrusion rate rather than improving them (Appendix C). The intervention signal is computationally usable, but it is not itself a planning policy. A responsive human–robot study and a policy designed on separate development data are still needed for a downstream safety claim.

6. Discussion

6.1. What the safeguard adds beyond a neural predictor

The graph ensemble should remain unmodified when its members agree: equivalence tests and low familiar firing rates support that choice. The safeguard matters when a topology change produces a small set of member-specific failures. In the crossing stress set, targeted routing improves the high-spread tail beyond fixed or permuted blending. More importantly, the same frozen rule improves the tail of the untouched T-junction. This is a concrete value not supplied by the neural mean alone: a causal action for detectable instability without waiting for labels or retraining the predictor.

The results also explain why the gains are not larger. The HGB and delayed-error gate can beat GEL-Ped in the crossing mean because much of that regime favours the alternative expert. Yet most neural error is shared across members. No function of same-family spread can expose that shared bias. The appropriate comparison is therefore not whether GEL-Ped replaces a strong predictor or online adaptation; it is whether the safeguard identifies a recoverable tail while preserving ordinary operation. The fixed and permuted controls, error decomposition, retraining study, second backbone, and untouched confirmation jointly support that narrower claim.

6.2. Use and limitations

GEL-Ped is intended for a reactive prediction layer. It may supply both $\hat{\mathbf{v}}_i$ and g_i to a local controller, monitoring service, or adaptation trigger. The replay shows that a simple inverse-confidence margin is not automatically useful; mapping g_i to control must be designed and validated as its own policy. The method is not a multi-second trajectory generator and does not replace destination modelling, multimodal forecasting, or reciprocal simulation.

Other limitations are substantive. The primary models use trajectory coordinates rather than gaze, pose, scene images, group identity, or explicit destinations. Calibration comes from seven corridor runs, and the first origin of a new track has less temporal support. The untouched confirmation contains 11 runs from one experimental family, not mixed traffic. The fallback is a feature-based regressor, not a mechanistic theory. Finally, GEL-Ped does not provide valid prediction intervals under shift. Adaptive conformal inference is a promising separate extension (Gibbs and Candès, 2021; Dixit et al., 2023), but no coverage guarantee is claimed here.

7. Conclusion

GEL-Ped turns ensemble disagreement into a simple runtime action for short-horizon pedestrian prediction. The neural mean remains unchanged while local spread lies inside a complete-run calibration envelope. Beyond that envelope, the forecast moves smoothly toward a compact, differently structured fallback.

The main result is not a universal gain in average accuracy. It is a replicated and independently confirmed reduction in the error tail associated with member-specific instability, obtained while familiar and mildly altered corridors remain practically equivalent to the graph ensemble. Fixed and permuted controls show that the effect is targeted; retraining and a Transformer check show that it is not tied to one seed or one encoder; and the untouched T-junction confirms the tail benefit under a new layout. The decomposition is equally important: most error under flow change is shared bias, outside the reach of same-family disagreement. GEL-Ped is therefore a focused safeguard with an explicit operating boundary, not a substitute for broader training, adaptation, or safety validation.

Data availability

The bidirectional-corridor trajectories are available at DOI 10.34735/ped.2013.5, the perpendicular-crossing trajectories at DOI 10.34735/ped.2013.4, and the T-junction confirmation at DOI 10.26165/JUELICH-DATA/YNCHYF. Raw files are not redistributed.

Code availability

Code, immutable run roles, the one-shot registry, compact result tables, and regeneration scripts are available at github.com/AmirNetwork/GEL-Ped. The submission snapshot records package versions, random seeds, checksums, and the commands used for every table and figure. Large per-forecast files are released separately with checksums rather than embedded in the manuscript archive.

CRediT authorship contribution statement

Amir Ghorbani: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review and editing.

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Declaration of generative AI and AI-assisted technologies

During preparation, the author used OpenAI Codex for language editing, code review, and figure-production assistance. The author reviewed and edited all outputs and takes full responsibility for the manuscript.

Ethics statement

This study reuses public, de-identified pedestrian trajectories and does not access participant identities.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Appendix A. Data roles and frozen analysis record

Table A.1: Complete-run roles. The crossing set was visible during method development; the T-junction was opened for labelled analysis once after the full pipeline was frozen.

Role	Runs	Native rate	Use
Corridor calibration	7	25 Hz	fitting, out-of-fold envelope, model selection
Familiar corridor	5	25 Hz	locked in-support equivalence
Altered corridor	3	25 Hz	locked geometry change
Right-angle crossing	13	25 Hz	development-informed stress and mechanism analysis
T-junction	11	16 Hz	one-shot untouched topology confirmation

The archive checksum and inventory are recorded in the frozen registry. The pre-outcome release tag is `gelped-v1.0-preheldout`. Prediction was executed without targets and produced a signed receipt before the confirmatory analysis read labels. The analysis script then wrote a single completion marker and refuses to overwrite it. Complete-run results, rather than individual origins, are the inferential units.

Calibration uses all chronological out-of-fold scores. The primary threshold and scale are not tuned on the stress or held-out regimes. Because deployed members are fitted on all seven calibration runs, their familiar firing rate (2.0%) is lower than the out-of-fold nominal exceedance; this realised rate is reported rather than assumed to be 10%. The 1.6-s window is computed before any 6,000-origin training cap is applied.

Appendix B. Retraining and backbone robustness

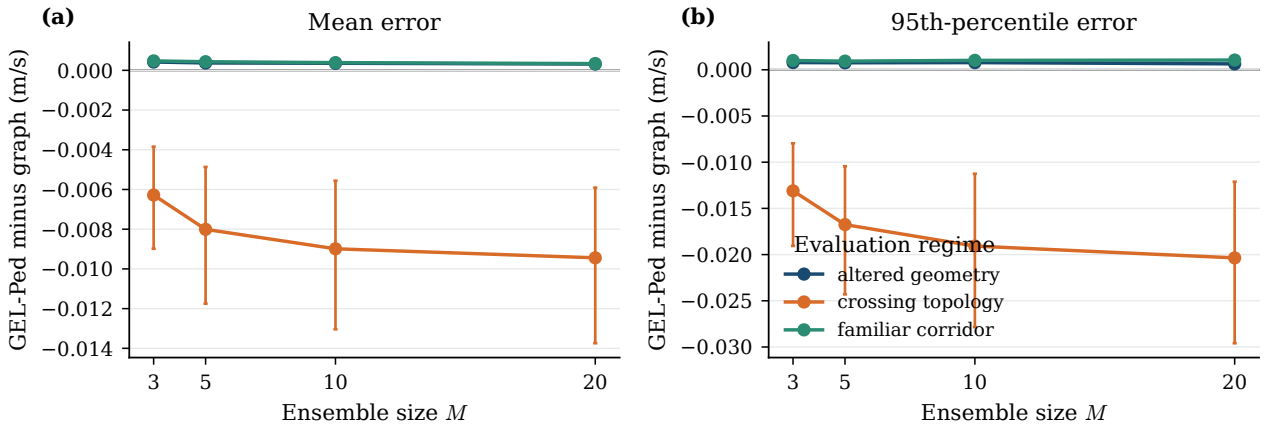


Figure B.1: Ensemble-size and member-draw stability. Points are run–draw macro effects and bars are crossed 95% resampling intervals. Negative values favour GEL-Ped. All 30 five-member draws improve crossing RMSE; the compact numerical results are released with the source.

The crossed intervals resample physical runs and complete member draws. Overlapping member subsets are not treated as independent model-development experiments. The Transformer uses standardised inputs, early stopping on a held-out complete run, and weight averaging. These checks support transfer of the guard mechanism across the two tested compact backbones, not a claim about all trajectory architectures.

Table B.1: Backbone generalisation: the frozen guard applied to a compact interaction Transformer trained and calibrated under the same run-wise protocol.

Regime	Method	Runs	RMSE	$q_{.95}$	$\mathrm{CVaR}_{.95}$	Firing
altered geometry	Transformer ensemble	3	0.1841	0.3392	0.4116	4.6%
altered geometry	HGB fallback	3	0.1903	0.3496	0.4214	4.6%
altered geometry	Guarded Transformer	3	0.1842	0.3392	0.4116	4.6%
crossing topology	Transformer ensemble	13	0.3767	0.6822	0.8471	41.5%
crossing topology	HGB fallback	13	0.3646	0.6602	0.8294	41.5%
crossing topology	Guarded Transformer	13	0.3699	0.6694	0.8347	41.5%
familiar corridor	Transformer ensemble	5	0.1888	0.3447	0.4230	4.0%
familiar corridor	HGB fallback	5	0.2004	0.3649	0.4445	4.0%
familiar corridor	Guarded Transformer	5	0.1891	0.3452	0.4239	4.0%

All velocity-error quantities are in m/s.

Appendix C. Operational audit and implementation footprint

Table C.1: Paired reactive-planner replay on identical recorded crowd motion and routes. The replay tests operational consequence, not reciprocal human response.

Regime	Policy	Complete	Travel (s)	Intrusion rate	Min. sep. (m)	Stop rate
altered geometry	Direct + fixed margin	100%	3.47	0.541	0.183	0.276
altered geometry	GEL-Ped + fixed margin	100%	3.87	0.596	0.230	0.397
altered geometry	GEL-Ped + adaptive margin	100%	2.53	0.419	0.172	0.467
crossing topology	Direct + fixed margin	100%	1.78	0.144	0.497	0.158
crossing topology	GEL-Ped + fixed margin	100%	1.82	0.144	0.497	0.173
crossing topology	GEL-Ped + adaptive margin	100%	2.49	0.207	0.394	0.356
familiar corridor	Direct + fixed margin	100%	1.60	0.217	0.486	0.167
familiar corridor	GEL-Ped + fixed margin	100%	1.60	0.217	0.486	0.167
familiar corridor	GEL-Ped + adaptive margin	100%	1.68	0.217	0.488	0.217

Travel time is averaged over completed replays. Intrusions use the predeclared 0.55-m personal-space threshold.

Table C.2: Implementation footprint and latency. Parameter counts and timings are kept in separate rows because saved-array timing excludes model forward passes.

Component	Parameters	Median (ms)	$p_{.95}$ (ms)	Scope
Graph member	13,203	–	–	Trainable weights
5-member graph ensemble	66,015	–	–	No weight sharing
Transformer member	3,442	–	–	Trainable weights
5-member Transformer ensemble	17,210	–	–	No weight sharing
five-member reduction	–	0.006	0.007	Post-prediction CPU
instantaneous guard and blend	–	0.028	0.032	Post-prediction CPU
reactive search with direct forecast	–	0.161	0.205	Post-prediction CPU
reactive search with guarded adaptive forecast	–	0.162	0.210	Post-prediction CPU

Latency is the median across complete-run timing records for each stage; each record uses its declared warm-up and repeated calls.

Each replay starts from the same route and uses identical recorded pedestrian motion across policies. "Intrusion rate" is the share of planner steps with separation below the pre-stated 0.55-m personal-space threshold. Stop rate is the fraction of steps choosing the zero-velocity action. The adaptive-margin rule was frozen before these outputs were inspected. Its adverse crossing result is retained because it shows that a prediction intervention and a control policy are separate design problems.

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